

DES (Simplified XOR Demonstration)

1. Aim

To implement a simplified demonstration of the Data Encryption Standard (DES) using a single XOR operation, and to understand how a symmetric key is used to encrypt and decrypt data.

2. Algorithm

1. Start the program.
2. Declare integer variables for plaintext, key and cipher.
3. Read the plaintext value entered by the user.
4. Read the key value entered by the user.
5. Compute the cipher value by performing the XOR operation between plaintext and key.
6. Display the encrypted (cipher) value.
7. Decrypt the cipher by performing the XOR operation between cipher and the same key.
8. Display the decrypted value and stop the program.

3. Program

```
#include <stdio.h>

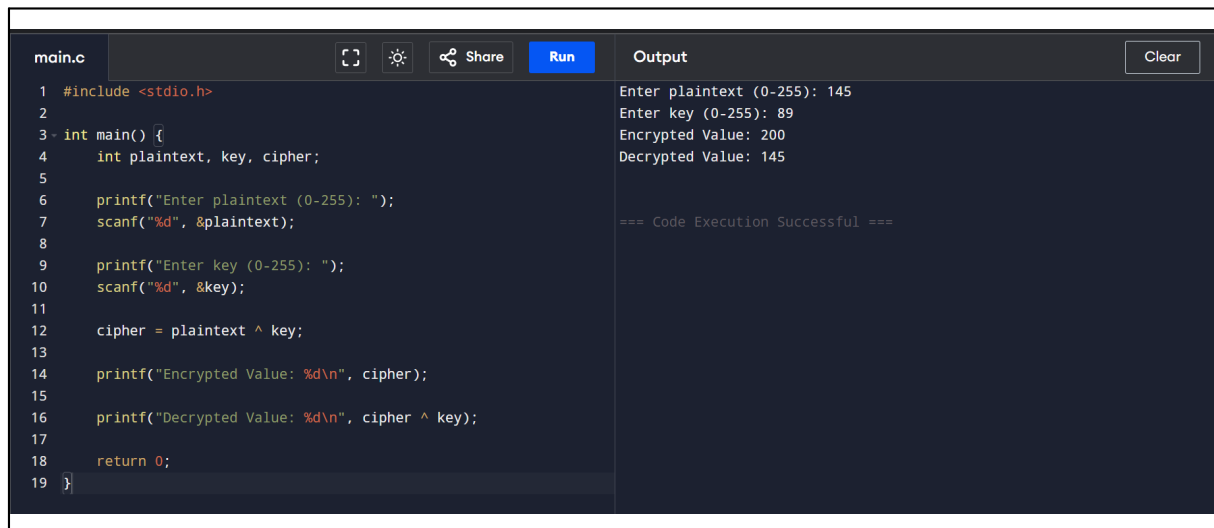
int main() {
    int plaintext, key, cipher;

    printf("Enter plaintext (0-255): ");
    scanf("%d", &plaintext);
    printf("Enter key (0-255): ");
    scanf("%d", &key);

    cipher = plaintext ^ key;
    printf("Encrypted Value: %d\n", cipher);
    printf("Decrypted Value: %d\n", cipher ^ key);

    return 0;
}
```

4. Output Screenshot



The screenshot shows a code editor with a file named 'main.c'. The code implements a simplified DES encryption and decryption program using XOR. It prompts the user for a plaintext value (0-255) and a key value (0-255). The program calculates the cipher as $\text{plaintext} \oplus \text{key}$ and then decrypts it back to the original plaintext as $\text{cipher} \oplus \text{key}$. The output window shows the execution results for the input values 145 and 89, resulting in an encrypted value of 200 and a decrypted value of 145. The execution was successful.

```
1 #include <stdio.h>
2
3 int main() {
4     int plaintext, key, cipher;
5
6     printf("Enter plaintext (0-255): ");
7     scanf("%d", &plaintext);
8
9     printf("Enter key (0-255): ");
10    scanf("%d", &key);
11
12    cipher = plaintext ^ key;
13
14    printf("Encrypted Value: %d\n", cipher);
15
16    printf("Decrypted Value: %d\n", cipher ^ key);
17
18    return 0;
19 }
```

Output:

```
Enter plaintext (0-255): 145
Enter key (0-255): 89
Encrypted Value: 200
Decrypted Value: 145

=== Code Execution Successful ===
```

5. Sample Input / Output

```
Enter plaintext (0-255): 65
Enter key (0-255): 15
Encrypted Value: 78
Decrypted Value: 65
```

6. Result

The simplified DES program using XOR-based encryption and decryption was implemented and executed successfully, and the original plaintext was correctly recovered after decryption.